

3.8.6 Maximum-likelihood parameter estimation Example

Q: 200 sample 2 class

$$\mu_1 = [-0.1055 \quad -0.0974]^T$$

$$\mu_2 = [2.0638 \quad 3.0451]^T$$

$$C_1 = \begin{bmatrix} 1.0253 & -0.0036 \\ -0.0036 & 0.8880 \end{bmatrix}$$

$$C_2 = \begin{bmatrix} 1.1884 & -0.013 \\ -0.013 & 1.0198 \end{bmatrix}$$

$$p(x|w) = \frac{1}{(2\pi)^{d/2} |\hat{C}|^{1/2}} e^{-\frac{1}{2}(x-\hat{\mu})^T \hat{C}^{-1}(x-\hat{\mu})}$$

$$g(x) = p(x|w) p(w)$$

Solution ①

$$p(x|w_1) = \frac{1}{(2\pi)^{\frac{2}{2}} \sqrt{|C_1|}} e^{-\frac{1}{2} (x - \hat{\mu}_1)^T C_1^{-1} (x - \hat{\mu}_1)}$$

$$= \frac{1}{2\pi \sqrt{1.0253 \times 0.888 - 0.0036 \times 0.0036}} e^{-\frac{1}{2} (x - \begin{bmatrix} -0.1055 \\ 0.0923 \end{bmatrix})^T C_1^{-1} (x - \begin{bmatrix} -0.1055 \\ 0.0923 \end{bmatrix})} \dots$$

$$= \frac{1}{2\pi \times 0.9542} e^{-\dots}$$

② $p(x|w_2)$

③ $g_1(x) = \frac{100}{200} \times p(x|w_1)$

$$g_2(x) = \frac{100}{200} \times p(x|w_2)$$

④ $g_1(x) > g_2(x) \rightarrow \text{class 1}$

$$g_2(x) > g_1(x) \rightarrow \text{class 2}$$

$$g_1(x) = g_2(x) = \text{decision boundary}$$